Sahaj Bhadja

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Summary

Software Engineer pursuing an M.S. in Computer Science at USC with 2+ years building **consumer-facing full-stack products**, **high-traffic APIs**, and **AI-assisted development workflows** for scalable, localized, and data-driven user experiences.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built **Python** workflow services and user-facing data interfaces for **500+ users**, reducing latency from **1.2s to 350ms** through query optimization while improving reliability in high-traffic product workflows.
- Implemented fault-tolerant orchestration for **5K+ monthly jobs** with validation safeguards and rollback checks, improving **production stability** and API reliability across distributed systems with real-time user impact.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React** and **Flask** features handling **10K+ daily requests**, shipping consumer-friendly product surfaces and secure backend APIs through fast prototype-to-production iteration with cross-functional collaboration.
- Improved release velocity with **CI/CD** controls and concurrent processing, reducing turnaround time by **60%** while preserving quality through automated checks, experimentation loops, and maintainable test practices.

USC AI Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python** experimentation pipelines over **100K+ trajectory logs**, enabling data-driven analysis and reproducible evaluation that informed product iteration, recommendation logic tuning, and user-behavior insights.
- Produced technical summaries and diagnostics that translated findings into implementation plans, improving **cross-functional collaboration** and accelerating decision-making across engineering, product, and research stakeholders.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API** and **ETL** microservice workflows with OOP, AWS Glue, and S3, improving response times by **30%** while supporting scalable integrations and dependable data exchange across distributed services.
- Maintained **20+** unit and integration suites using Jenkins, PyTest, and Mockito across **Linux/Windows** environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%** via automation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **REST API** and **WebSocket** modules supporting eight workflows and **2M+ synchronized records**, improving responsiveness and reliability in user-facing, transaction-heavy application workflows.
- Automated deployments and testing with **Docker** and **Jenkins**, reducing deployment time by **50%** and improving release consistency through scripted validations and robust rollback controls.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Bash
Frontend: React, TypeScript, JavaScript, Responsive Web UX, Consumer Product Interfaces
Backend & APIs: Flask, FastAPI, REST APIs, Microservices, WebSockets, Distributed Services
Cloud & Scale: AWS (S3, Glue, ECS), Docker, Kubernetes, Azure, Linux/UNIX, Monitoring
Engineering Practices: CI/CD, Unit/Integration Testing, Experimentation Mindset, Git, Jenkins, Agile/Scrum

Projects

FinNexus | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*

- Built a scalable consumer fintech platform supporting **50K+ users** and **1M+ transactions** with secure integrations, real-time updates, and localized, high-availability workflows for global users.

PiXMod | *Next.js, CLIP, SAM, Stable Diffusion, MongoDB*

- Built an AI visual editing product using **SAM** segmentation and **Stable Diffusion**, processing **1K+ assets** while improving interactive UX quality and rapid experimentation for user-facing features.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized **RESTful API** on AWS CodePipeline and ECS/Fargate with automated **CI/CD** and validation checks, achieving **87% accuracy** and production-grade reliability under sustained workloads.